

Immunofluorescence staining for cells

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Protocol adapted from Rotem Ben-Tov Perry (Ulitsky Group, WIS)

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No need to rock/shake plate

Day 1

1. If working with Ibidi Chambers, prepare a humid chamber by placing Whattmann paper inside a large Petri dish, wet the paper and place parafilm on it. If working with adherent cells, this step is not necessary.
2. For neurons: Treat with DNase I/RNase first
3. Wash cells once with 1X PBS at room temperature (RT). – for neurons don't do this
Use 500 ul volume per 24well when working with coverslips inside the 24well and use P1000 for neurons (no vacuum)
For other cells can use serological pipettes
For neurons: prepare 8%PFA and mix 1:1 with media in the well. Use fresh PFA always for neurons.
4. Fix cells for 15 minutes with cold 4% PFA in 1X PBS at RT inside a chemical cabinet.
5. Wash twice for 5min with 1X PBS at RT.
6. Add the Triton Permeabilization Solution and incubate for 30min at RT. Do not incubate for more than 30min!
7. Wash three times for 5min with 1X PBS at RT.
8. Dilute the primary antibody/antibodies in Blocking Solution. Incubate the primary antibodies ON at 4°C (in the fridge, standing)
Combine antibodies according to host species. Prepare one tube per combination of antibodies.

Day 2 (all steps are performed in the dark and/or using aluminum foil as cover)

1. Wash three times for 5min with 1X PBS at RT.
For neurons first wash with PBS with 0.1% Triton
2. Dilute the secondary antibody/antibodies in Blocking Solution. Incubate the secondary antibodies for 2h at RT, protected from light.
For neurons add DAPI at this point 1:1000.
3. Wash three times for 5min with 1X PBS at RT.
For neurons first wash with PBS with 0.1% Triton
4. For other cells: Dilute DAPI 1:5.000 in 1X PBS, and incubate 10min at RT.
5. Wash once for 5min with 1X PBS at RT.
6. Final mounting:

For standard IF, you can dilute prolong glass 1:2 into ultrapure H₂O.For neurons do not dilute the mounting media.

- a. Coverslips/Ibidi Chambers: Remove excess PBS using tissue paper or Whattmann paper. Add enough Prolong Glass/equivalent over the cells and carefully place the coverslip over the cells.
- b. Adherent cells inside wells: Add fresh 1X PBS and store at 4°C.

Triton Permeabilization Solution

Reagent	1ml	3ml	5ml
100% Horse Serum	50ul	150ul	250ul
50mg/ml BSA	20ul	60ul	100ul
10% Triton	10ul	30ul	50ul
1X PBS	920ul	2760ul	4600ul

Blocking Solution

Reagent	1ml	3ml	5ml
100% Horse Serum	50ul	150ul	250ul
50mg/ml BSA	20ul	60ul	100ul
1X PBS	930ul	2790ul	4650ul

Antibodies used:

Target	Brand	Dilution	Host
Nanog	Cell Signaling	1:250, 1:500	
Oct4	Cell Signaling	1:250, 1:500	
Map2	Cell Signaling	1:500	mouse
Tubulinbeta	Abcam	1:1000	
Cux1	Abcam	1:500	
GR	Cell Signaling	1:500	rabbit
GFP		1:500	Rat
anti-rabbit AF488		1:1000	Goat
Anti-mouse AF 555		1:1000	Goat
Anti-rat AF488		1:1000	